



Superstore Insights: Data-Driven Business Strategy

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Dashboard Preview

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Total Sales by Region	Total Profit by Region
🟪 West — 725.46K (31.58%)	🟪 West — 108K
🟡 East — 678.78K (29.55%)	🟡 East — 92K
🟢 Central — 501.24K (21.82%)	🟠 South — 47K
🟠 South — 391.72K (17.05%)	🟢 Central — 40K



Interactive dashboards built in Power BI — see </dashboard/> folder for **.pbix** file

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About

Superstore Insights is an end-to-end data analytics project built on the classic Sample Superstore retail dataset. It combines **MySQL** for structured querying and **Power BI** for interactive visualization to uncover actionable business intelligence from over 9,900 retail transactions across the United States.

This project demonstrates the full analytics pipeline — from raw data ingestion to strategic business recommendations — making it a complete showcase of real-world data analysis skills.



Purpose

The core objective of this project is to answer a fundamental business question:

"Where is the Superstore making money, where is it losing money — and why?"

By slicing data across **regions, categories, shipping modes, customer segments, and discount structures**, this project surfaces the hidden inefficiencies and overlooked opportunities that drive or drain profitability. The goal is not just analysis — it's to equip business decision-makers with the insight to act.



Dataset Overview

Propert	Detail
Source	Sample Superstore (Tableau / Kaggle public dataset)
Records	~9,994 rows
Time Period	4 years of US retail transactions
Format	CSV → imported into MySQL
Key Fields	Ship_Mode, Segment, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit

Regions Covered: West · East · Central · South

Product Categories: Furniture · Office Supplies · Technology

Customer Segments: Consumer · Corporate · Home Office



Approach

The project was executed in three phases:

1. Database Setup (MySQL)

- Created a dedicated `superstore_project` database
- Designed a `sales` table with appropriate data types (`DECIMAL`, `VARCHAR`, `INT`)
- Imported and verified 9,994 rows of clean retail data

2. Exploratory & Deep-Dive SQL Analysis

- Wrote 10+ targeted SQL queries covering revenue, profit, discounts, shipping, and segments
- Used `GROUP BY`, `HAVING`, `ORDER BY`, `NULLIF`, `ROUND`, and aggregate functions
- Identified loss-making sub-categories, high-discount traps, and top-performing geographies

3. Power BI Visualization

- Connected Power BI to query outputs
- Built interactive charts: donut chart for regional sales share, bar charts for profit comparison
- Applied consistent color coding by region for clarity

Questions Solved in the SQL File

The `superstore_analysis.sql` file is structured around 10 analytical blocks:

#	Query Block	What It Does
1	Database & Table Setup	Creates schema, defines column types, imports data
2	Data Verification	<code>SELECT *</code> and <code>COUNT(*)</code> to confirm successful import
3	Overall KPIs	Total revenue, total profit, average discount
4	Category Performance	Revenue and profit breakdown by product category
5	Loss-Making Sub-Categories	Filters sub-categories where <code>SUM(profit) < 0</code>
6	Top 5 Most Profitable Cities	Best performing cities by total profit
7	5 Least Profitable Products	Sub-categories with the lowest profit margins
8	High Discount Analysis	Avg discount % vs total profit by category
9	Shipping Mode Performance	Order count, total sales, avg profit per shipping method
10	Customer Segment Deep Dive	Revenue, profit, and profit margin % by segment
11	High Sales / High Loss Mystery	Identifies sub-categories that sell well but lose money
12	Regional Performance Deep Dive	Cities reached, sales, profit, and margin % by region

? Key Questions Answered

 Which region generates the most revenue and profit?

West leads in both sales (\$725K) and profit (\$108K), followed by East (\$678K sales, \$92K profit). The Central region trails significantly in profit despite decent sales volume — signalling an efficiency gap.

 Which sub-categories are losing money?

Tables and **Bookcases** are the primary loss-makers in the Furniture category. These items are frequently sold at steep discounts, completely eroding their margins.

 Which cities are most profitable?

The top 5 most profitable cities are concentrated in the **West and East** regions, confirming their dominance as growth markets.

 Which shipping mode performs best?

Same Day shipping delivers the highest average profit per order, while **Standard Class** handles the most volume but at lower per-order profitability.

 Which customer segment is most efficient?

The **Home Office** segment achieves the highest profit margin percentage, making it the most efficient segment despite not being the largest by revenue.

🔥 Why do high-sales products sometimes lose money?

The discount analysis reveals that products like Tables have average discounts exceeding 30%, which turns high-sales volume into net losses — the classic "sell more, lose more" trap.

← END Conclusion

This project confirms that **revenue and profitability do not always move together**. The Superstore's biggest challenge isn't sales volume — it's **margin erosion through excessive discounting**, particularly in the Furniture category. Meanwhile, regions like the West consistently outperform not just in sales, but in operational efficiency.

The combination of SQL for precision querying and Power BI for visual storytelling proved highly effective in transforming raw transaction data into a clear, decision-ready business narrative.

💡 Recommendations

Based on the analysis, here are the top strategic recommendations:

Priori	Recommendation	Impact
High	Cap discounts on Tables and Bookcases at 10% — current discounts exceed 30% and cause direct losses	Immediate profit recovery
High	Audit the Central region — high sales, low profit margin signals operational inefficiency or pricing issues	Margin improvement
Medium	Double down on the West region — highest sales AND profit margin; ideal for expansion investment	Revenue growth
Medium	Prioritize Home Office segment marketing — highest profit margin %; underserved relative to its efficiency	Scalable growth
Low	Evaluate Same Day shipping pricing — high avg profit per order; consider unselling it more actively	Incremental gains
Low	Investigate Technology sub-category opportunities — strong margins; potential for bundling with loss-making Furniture items	Portfolio optimization

💬 Feedback and Suggestions

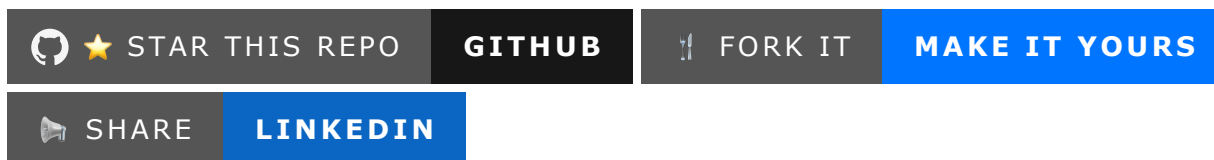
Have ideas to improve this analysis? Found an interesting pattern in the data I missed?

- 🐛 Found a bug or error? Open an [Issue](#)
- 💡 Have a feature suggestion? Start a [Discussion](#)
- 🤝 Want to contribute? Fork the repo, make your changes, and submit a Pull Request
- ✉️ Direct feedback? Reach me at [your.email@example.com]

All feedback is genuinely welcome — this project is a learning journey and every perspective helps!

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If this project helped you learn something new, inspired your own analysis, or saved you time — consider showing your support:



- ★ **Star the repository** — it helps others discover this project
- 🍴 **Fork and build on it** — extend the analysis with your own queries
- 📢 **Share it** — tag me on LinkedIn if you use this as inspiration for your own project

"Data is the new oil — but only if you know how to refine it."

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